Conditional generation of visible language

a Training

60 visible glyph images
Frozen causal reader

→

Continuous visual history
384 state + 1024 image field

History conditions every residual scale

Four noisy future cells
Independent or shared noise levels

→

Conditional strip denoiser
FiLM and cell attention

→

Velocity and visual losses
5.90M trainable parameters

b Inference

Noise and visible history

→

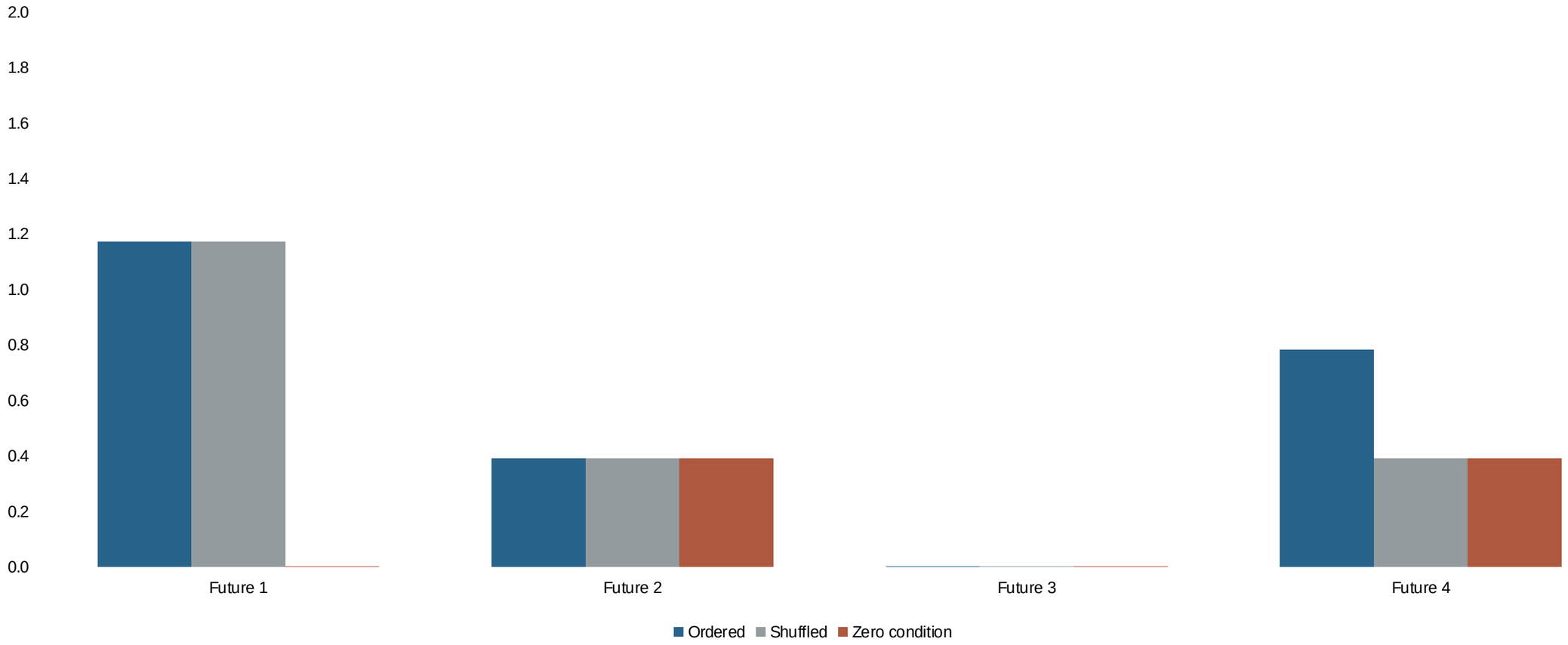
32-level reverse process

→

Four output images
Append pixels to next context

Source: V53 frozen protocol, implementation and locked audit. Schematic objects are editable.

Direct block generation under paired visual histories



256 prefixes, one primary sample each, four output cells
Ordered block identity 0.586%, shuffled 0.488%, zero condition 0.195%

Source: locked V53 development report. Native chart includes its editable evidence workbook.

Unselected output images expose the language error

Fixed first eight audit rows. Pixels are model output, with nearest-neighbour enlargement. No glyph replacement.

Exact raster likelihood preserves a visible copy path

64 writing images
Frozen 16.28M reader

→

Continuous visual state
Hidden384 + image field1024

→

Fixed inverse field
Aligned signed pixel plan

Training: exact Bernoulli image NLL

Earlier target rows
Four-layer row transformer

→

Earlier pixels within row
Strictly masked MADE

→

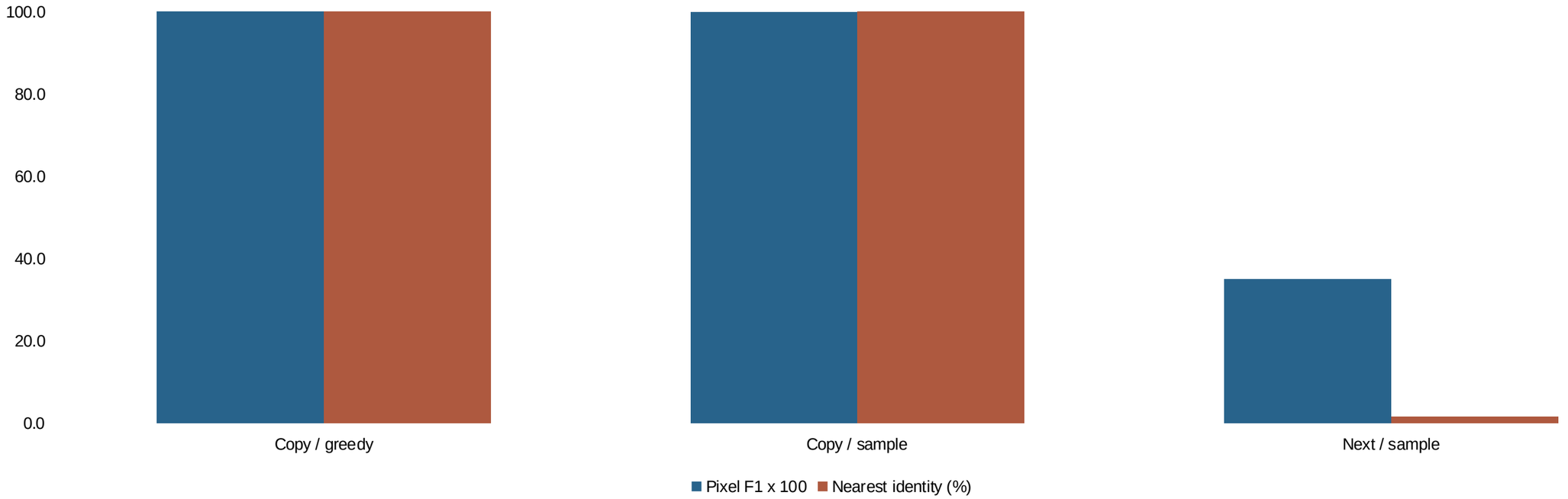
Aligned logit + residual
Normalized next-image law

25% supplied-image motor exercises; 75% natural predicted conditions. No glyph inventory.

Inference: sample 1,024 causal pixels; append the visible raster to history

2.43M new density parameters; 18.71M total. Supplied-image copying is architectural, not language evidence.

Copying succeeds; autonomous language remains unresolved

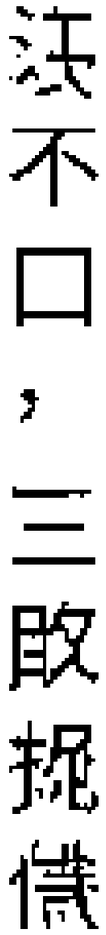


Exact-image copying: 256/256 greedy. Autonomous target identity: 4/256 ordered, 3/256 shuffled.

Target NLL: 15.219 ordered; 15.254 shuffled nats/image
Paired advantage 0.0355 nat; source-record bootstrap 95% CI [-0.1014, 0.1693]

V54 locked development audit. Native chart and embedded workbook. Supplied-image controls reveal the answer.

Actual rasters: exact motor copying and failed continuation

Target	Supplied / greedy	Ordered / sample	Shuffled / sample	Ordered / greedy
				

First eight fixed audit rows; nearest-neighbour enlargement. No OCR, image replacement or selected examples.